

SOWKATH ALI L

Software Developer

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PROFESSIONAL SUMMARY

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Computer Science and Engineering undergraduate with hands-on experience in software development, backend API development, databases, and AI-based applications. Experienced in building software solutions using **React.js, Node.js, Express.js, Python, FastAPI, MySQL, MongoDB, and REST APIs**. Developed real-world projects including an **AI-driven emergency response system, an intelligent land record digitization platform, and AI-based cybersecurity solutions**. Strong interest in software engineering, backend development, and AI/ML, with a focus on building practical and scalable solutions to real-world problems.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL

Frontend: React.js, HTML5, CSS3

Backend: Spring Boot, Spring MVC, Node.js, Express.js, FastAPI, REST APIs

Databases: MySQL, MongoDB, SQLite

ORM & Persistence: Spring Data JPA, Hibernate

Security: Spring Security, JWT

AI/ML: scikit-learn, OpenAI API, Generative AI

Tools: Git, GitHub, Postman, Docker

Core Concepts: OOP, Data Structures & Algorithms, DBMS, REST Architecture

Soft Skills: Problem Solving, Teamwork, Adaptability, Quick Learning

INTERNSHIP EXPERIENCE

MERN Stack Developer Intern — NoviTech R&D Private Limited *March 2026 – June 2026 (3 Months)*

- Developed responsive web applications using the MERN Stack.
- Built frontend interfaces using React.js, HTML, CSS, and JavaScript.
- Developed backend APIs using Node.js and Express.js, and worked with MongoDB for database operations.
- Integrated frontend and backend using REST APIs; participated in debugging, testing, and version control with Git.

Full Stack Developer Intern — NoviTech R&D Pvt. Ltd. *March 2026 – April 2026 (1 Month)*

- Assisted in building and testing full stack features across the MERN stack.

UI/UX Design Developer Intern — NoviTech R&D Pvt. Ltd. *March 2026 – April 2026 (1 Month)*

- Designed and refined user interfaces and improved usability for internal projects.

PROJECTS

Intelligent Land Record Digitization & Validation System (DLR)

Tech Stack: React.js, Node.js, Express.js, MySQL, Python, FastAPI, AI/OCR, REST APIs

- Developing an AI-powered land-record platform to digitize and validate scanned, handwritten, and legacy land records and convert unstructured documents into structured information.
- Designed role-based workflows for Citizens, Government Officers, Ministers, and Super Admin, with secure authentication and dashboard-based access.
- Implemented a full-stack architecture connecting a React-based frontend with Node.js/Express REST APIs and a MySQL database.

- Designed an AI service using Python/FastAPI for document processing, OCR, information extraction, and validation workflows.
- Built workflows for landowner details, survey numbers, land area, location, land classification, and other land-record information.
- Designed the system to support multilingual document processing, validation, GIS/map-based land visualization, and digital government services.

AI-Driven Emergency Response System (MERN Stack)

Tech stack: React.js, HTML5, CSS3, JavaScript, Node.js, Express.js, MongoDB

- Built a software-based emergency response system that detects vehicle accidents, captures GPS location, assesses severity using rule-based logic, and coordinates hospitals, ambulances, and police in real time.
- Created separate dashboards for users, hospitals, ambulances, and police.
- Designed as a software simulation with scope for future integration with real vehicle sensors and AI-based severity prediction.

AI Chatbot for Wheat Farmers

Tech stack: Node.js, Express.js, SQLite, OpenAI API

- Built a conversational AI assistant that answers wheat farming questions for Indian farmers using the OpenAI API.
- Implemented a Node.js/Express backend with a REST endpoint that processes user queries and returns AI-generated responses.
- Used SQLite to log and persist chat history for each conversation.

AI-Based Adaptive Firewall

Tech stack: Python, scikit-learn, Isolation Forest, iptables

- Developed a network security tool that scores incoming traffic events for risk using a trained classifier combined with an Isolation Forest anomaly detector.
- Implemented automatic IP blocking/unblocking with a configurable time-based blacklist and a dry-run mode for safe testing.
- Designed feature extraction from network event data (bytes, packets, duration, ports, protocol) to feed the risk-scoring model.

EDUCATION

Bachelor of Engineering — Computer Science and Engineering

Academic year 2023-27

CGPA: 8.0/10 (up to Semester VI) Annapoorana Engineering College (autonomous), Salem, Tamil Nadu

Higher Secondary Education (12th) — State Board of Tamil Nadu

2023

51% Gokulanatha Hindu Mahajana Boys Hr Sec School, Salem

SSLC (10th) — State Board of Tamil Nadu

2020

49.6% The Gugai Hr Sec School, Gugai, Salem

CERTIFICATIONS

- MERN Stack Internship — NoviTech R&D
- Full Stack Dev Internship — NoviTech R&D
- UI/UX Design Internship — NoviTech R&D
- Power BI Micro Course
- Advanced Full Stack Web Development — softrixx Innovation

ACHIEVEMENTS

- Participant, India AI Impact Buildathon 2026 (GUVI-HCL)
- Paper Presentation, NCEPNCA-2025 — Automated Vehicle Speed and Helmet Compliance Detection using Deep Learning
- Participant, DCG Coimbatore Cybersecurity Chapter Meetup 2026
- Participant, Smart India Hackathon (SIH) 2025